

Incentives in The Gale-Shapley Stable Matching Algorithm

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Abstract

The Gale-Shapley algorithm (GS) and its extensions have well-established real-world applications. However, algorithm designers sometimes apply GS systematically, overlooking the broader institutional and performance-based environments in which matching takes place. Consider a situation where, prior to the GS matching process, participants compete in a marathon that grants top performers their preferred matches. What if the objective is to optimize both performance and preference-based matching? Existing hybrid algorithms either converge to the men-optimal GS outcome or lack formalization as a generalizable framework. In this paper, we examine such cases, which are common in college admission-type matching problems. We introduce an incentive mechanism that rewards high performers during the matching process while still maximizing preferences on both sides. Using a CES function, we formalize this framework and demonstrate how it can yield better overall outcomes compared to both the standard GS and purely performance-based systems. Drawing from a real-world example of a professional training school, we develop algorithms that enable planners to flexibly combine these two inputs to maximize global outcomes. We present applications in college admissions-type matching, maximization of sports teams' revenues, and we demonstrate that these new algorithms also hold potential in health policy, calling for further research in that area. Moreover, an application to inclusive development strategies illustrates how federal governments can leverage this theoretical framework to enhance the performance of local states in specific sectors, thereby achieving higher aggregate outcomes.

Keywords: Assignment, Gale-Shapley algorithm, Amphi-garrison, AH-GS model, Incentives, CES.
JEL : C78, D47, D61.

1 Introduction

The classic two-sided assignment problem introduced by [Gale and Shapley \(1962\)](#) involves two distinct sets, such as men and women, each with preference rankings over the members of the other group. The algorithm produces stable matchings where no pair of individuals would both prefer to be with each other rather than their assigned partners. Among all stable matchings, there exists an optimal matching, where every individual is at least as well off as they would be under any other stable matching. These algorithms generally promote global satisfaction and have inspired extensive research and application across various fields. However, the implementation of these algorithms overlooks sometimes the specific context and environment of certain problems, thereby reducing their practical relevance despite theoretical optimality. This issue arose, for instance in the France's National School of Administration (ENA)¹ when a [Decree \(2024\)](#) replaced the famous and historical purely performance-based system known as the "amphigarnison" (AH) by a two-side preferences matching related to the GS to assigned graduates to different departments. The abandoned AH model was giving high-performing students the first choice of assignments which becomes final. Some algorithmists are proponents of those abrupt shifts as the AH system is considered outdated and inefficient. This is the case for [Teo et al. \(1999\)](#) who State that the AH is plagued by several problems, proposing to replace it by the GS to assign primary school students in Singapore to secondary schools. Those methods are overlooking the objective of the AH which is to foster performance during training prior to the matching process. From Norman Triplett (1898), through [Zajonc \(1965\)](#), to more recent studies ([Konings and Hettinga, 2018](#)), we know that humans tend to perform better in the presence of competition.

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In case of ENA, since students know that those ranked higher will have better chances of being posted to their preferred locations, the system fosters strong competition during training, thereby improving student performance and professional efficiency which is the ultimate purpose of civil servants' schools. Replacing this with a matching algorithm like the GS might reduce student motivation during training, as assignments would be determined in advance based on both sides' preferences.

The AH system has also clear inefficiencies in this case as it lacks consideration for students' prior academic specializations. For example, a highly ranked medical graduate could choose a post in a finance institution, leading to mismatch between qualifications and job roles drifting away from the objective of that training school. Moreover, the AH yield in a contradiction between rigorous recruitment and training on one hand, and a loosely regulated final assignment on the other; raising concerns about the overall effectiveness of the system. These weaknesses of AH recall that assigning students based on their preferences and academic backgrounds would yield in more coherent and stable outcomes, even if it sacrifices some of the competitive incentives. This sets off a vicious cycle of criticisms bouncing between AH and GS, each responding to other, a cycle that can be broken by introducing a new model AH-GS. This hybrid algorithms combine strengths of both systems while minimizing their respective weaknesses. This provides to the planner an additional strategic option for decisions. The development of those algorithms are made on a global framework which allows applications in broad domains.

2 Literature Review

Gale and Shapley (1962) present a situation where colleges have to admit students, with both parties having preferences on the other. The challenge is to assign applicants to colleges fairly. Gale and Shapley represent this scenario with a marriage problem consisting of two equal-sized, finite, and nonempty sets of men and women, where each individual has a strict preference ordering over the other set. The goal is to achieve a matching that maximizes overall satisfaction. In the first stage, only agents of one set (typically men) propose to members of the opposite set. Women receiving multiple proposals retain highest-ranked proposal based on their respective preferences and reject others. In the second stage, rejected men propose to their second choice and women compare any new proposals with their current temporary match. This process known as "deferred acceptance" continues until a stable matching is reached. Authors noted that this process not only ensures a stable match but also the optimal assignment. The assignment in the GS model is men-optimal since the proposing side (men) gets their best possible outcome.

From that core model, scientists have developed extensions and real-world applications, especially in computer science and among game theory. The impact of this model and its extensions was so significant that the Nobel Prize in economics (2012) was awarded to Lloyd Shapley along with Alvin Roth who contributed as well in this domain.² One well-known application of the model is assignment of individual to institutions (such as college admissions, doctors to hospitals, ENA graduates to public service positions, etc.). These systems are widely used from decades in many country. For instance, in the United States, the National Residents Matching Program³ established in 1952, conduct an annual matching process of graduating medical students and residency program. A similar system exists in Canada (Canadian Resident Matching service).⁴

In the United States, the Boston public School among others, had adopted a different mechanism. Under this system, each student submits a ranked list of preferred colleges. Students are first proposed to their top-ranked institution, which retains the best candidates among those who apply. Rejected students are then proposed to their second choice, and so on. However, students who lose their ranking at top schools may lose priority in subsequent stages. Parents may even hide their children's true preferences, thinking it is not in their best interest to reveal them (Abdulkadiroğlu and Sönmez, 2003; Abdulkadiroğlu et al., 2005). Then, this process can lead to suboptimal allocation of school seats (Abdulkadiroğlu and Sönmez, 2003). Experimental study of Chen and Sönmez (2006) highlights some experimental deficiencies in the Boston system. Ergin and Sönmez (2006) showed that the GS has potentially more significant welfare gains than Boston system.

Significant applications of stable matching have also been developed in labor markets (Arcaute, 2009; Kelso Jr and Crawford, 1982) and in the market for kidney donation (the Nobel laureate A. Roth⁵ has published extensively on this subject) (Unver, 2004). In recent years the development of technology has allowed researchers to simulate and test their models more efficiently, and on the other hand, extended the

²<https://www.nobelprize.org/prizes/economic-sciences/2012/popular-information/>

³<https://www.nrmp.org>

⁴<https://www.carms.ca>

⁵<https://www.nobelprize.org/prizes/economic-sciences/2012/roth/lecture/>

applications of the algorithm. Shi et al. (2018a) highlighted recent applications such as assigning a large number of users to internet servers, matching peers in a P2P network according to preference, resource allocation in 5G networks, and the matching mechanism of real-world dating website.

In domain of public development planning, the China's Targeted Poverty Alleviation (TPA) offers relevant experience of how matching methods can improve country development strategies. Rolled out from 2013 to 2020, the program aimed to reduce inequality between poor and rich, urban and rural China as well (Zhang et al., 2020). Based on some indicators, the program matched wealthier or more resourceful entities (like departments, cities, or companies) with specific poor villages enabling those entities to directly support development and poverty reduction in those areas. Yang and Liu (2021) describe it as a match between the government-supported supply to rural poverty demand. Li and Zhou (2025) concluded that during the TPA period, the socioeconomic level of poor areas developed rapidly. See in the Appendix 6 how the assignment was tailored to the specific poverty root of each village.

3 A dual-variable optimization approach

Following developments are based on college admission matching type, especially of ENA which recruits and trains students before deploying them in public services. We consider a function $f(X, Y)$ representing the institutional impact. Since $X(a)$ and $Y(b)$ then $f(a, b)$. X represents the performance of ENA students during training and Y represents the quality of the final institutional match between graduates and the public institutions. We assume that all functions are positive with their exogenous variables. The incentive is not under study directly, hence we assume it has a positive impact. Also, we assume symmetric information : all agents are aware of the algorithm to be applied and respond accordingly.

Our scenario consists of : the outcome of the first game (represented by the final ranking of students), the incentive strategy (represented by the impact of the final ranking on the matching algorithm) and the matching process (defined through an algorithm such as Gale-Shapley that integrates the final ranking and both side preferences' considerations) . Thus, the planner seeks to maximize : $f(X, Y)$ subject to $X = X(a), Y = Y(b); a, b \in \mathbb{N}$, then $\max_{(a,b) \in \mathbb{R}_+^2} f(a, b)$.

Let determine a standard economic function for that scenario. The constant elasticity function introduced by Arrow et al. (1961) has been used in broad situations for optimization. Beyond the classic production and consumption representation, the CES has been used wide range of specific domains including energy resources allocation (Jo and Miftakhova, 2024), urban economics (Ahlfeldt and McMillen, 2022), trade policy (Bagwell and Lee, 2018; Baley et al., 2019) and agricultural policy (Czyżewski et al., 2019).

Theorem 1. *the function $f(X, Y)$ can be represented with the Constant Elasticity of Substitution (CES) function $f(X, Y) = (\alpha X^\rho + \beta Y^\rho)^{\frac{1}{\rho}}$ with $\alpha, \beta > 0$ and $\rho \in (0, 1)$*

Proof. the function $f(X, Y)$ has following characteristics matching with CES when $\rho \in (0, 1)$, :

1. Bivariate economic function. In our case $f(X, Y) : \mathbb{R}^2 \rightarrow \mathbb{R}$. To Maximize Institutional Impact of ENA Graduates, the planner aims :
 - Promoting performance through incentive-based competition : being aware that their final ranking will directly influence their institutional assignment, students work harder, improve their skills, and ultimately become more effective public servants. Such a mechanism aligns individual incentives with the institution's mission, *to strengthen public administration*.
 - Ensuring relevance through major-matching assignments: While ENA provides a solid foundation in public administration to all trainees, institutional performance tend to be optimal when a student's prior academic or professional background aligns with the major of the assigned institution. For instance, a doctor with ten years of training is generally better suited for health policy than someone with a finance background; and the reverse holds true. This enhances the efficiency and effectiveness of public institutions leading to ENA's objective: *improving public administration*.
2. Non-strict essentiality of inputs : the planner can decide to rely exclusively on AH or GS model and still find an outcome.

$$f(X, 0) = (\alpha X^\rho)^{\frac{1}{\rho}} = \alpha^{\frac{1}{\rho}} X \neq 0 \quad (1)$$

$$f(0, Y) = (\beta Y^\rho)^{\frac{1}{\rho}} = \beta^{\frac{1}{\rho}} Y \neq 0 \quad (2)$$

3. There is an inherent trade-off between maximizing either X or Y . The more the algorithm depends on X (ranking), the greater the motivation. The more it depends on Y (major alignment), the less the incentive is effective. Similar to the limited resources logic, the planner has constraints on levels of X and Y as demonstrated in next sections.
4. The function is strictly quasi-concave as demonstrated by the theorem 2. See CES in appendix 6

□

Corollary 1. *In our scenario $\rho \in (0, 1)$*

If $\rho \leq 0$ the characteristic 2 is not satisfied. If $\rho \geq 1$ the characteristic 4 is not satisfied. Then, the interval of p is $(-\infty, +\infty) \setminus (-\infty, 0] \cup [1, \infty) = (0, 1)$.

Theorem 2. *The hybrid model provides the largest range of control to the planner.*

$$f(X, Y) > \max \{f(X, 0), f(0, Y)\} \quad \forall X > 0, Y > 0$$

Proof. Let study the 3 possible cases of matching models in our scenario :

1. **AH Mechanism:** Students are ranked by training performance. The highest-ranked chooses first, followed by the next, and so on. In this case, the performance incentive is maximized and the planner can find $a = \arg \max X$. However this comes with Cons as the final matching depends entirely on students' preferences and the planner has no control over $\tilde{Y}$, which becomes a stochastic outcome. The planner policy can determine one certain outcome : $\alpha^{\frac{1}{\rho}} X(1)$ to be added to the stochastic outcome generated by $\tilde{Y}$.

Corollary 2. *Once X is maximized, no further control policy is possible on $\tilde{Y}$.*

In worst case, the $\tilde{Y}$ can lead to mismatches like “an accountant managing health policy.”

2. **GS Algorithm :** This uses the stable matching algorithm based on preferences of students and institutions. With GS, the matching based on preferences is optimized and the planner can find $b = \arg \max Y$. However, Incentives are removed since students know that their ranking does not matter, reducing motivation. Thus, the performance $\tilde{X}$ becomes a stochastic variable. The planner policy can determine one possible outcome : $\beta^{\frac{1}{\rho}} Y(2)$ to be added to the stochastic outcome generated by $\tilde{X}$.

Corollary 3. *Once Y is maximized, no further control policy is possible on $\tilde{X}$.*

3. **Hybrid Model (AH-GS):** Unlike for the two previous models, this combines both incentive and preferences matching optimization. The planner integrates ranking (to keep incentive) and domain relevance (to improve matching of academic backgrounds to institutions). Thereby, the planner has control over both X and Y ; the motivation is preserved, and major matching is achieved. However, the planner is not certain to obtain $a = \arg \max X$ and $b = \arg \max Y$ regarding the trade-off between variables.

The policy implication is clear : AH maximizes training performance but risks random poor matches. GS ensures good institutional matching but at the cost of motivation. AH-GS provides a balance, allowing the planner to retain influence over both motivation and assignment quality. The convexity assumption of the indifference curve makes this model interesting to analyze. □

4 Model properties

Let's formalize AH-GS model and provide its algorithms. The first merely integrates the ranking order in the GS model. In addition to the ranking order, the second integrates a variable which allows the planner to determine the incentive level.

4.1 Definitions

The followings are mathematical representations of the current ENA model. When a property breaks an assumption of classic Gale-Shapley model, we denote it by $\implies$ BGS. We consider two finite and nonempty sets U and E that refer to Institutions and ENA's students. $U = \{u_1, u_2, \dots, u_m\}$ and $E = \{e_1, e_2, \dots, e_n\}$. $|U|$ and $|E|$ are respectively cardinalities of U and E . Throughout the paper, we represent cardinality of each set the same way. Let r_i be the rank of student e_i ; $r_i \in \mathbb{N}_+^*$.

Remark 1. While agents in the set of institutions are ordered randomly, students in set E are ordered according to their final score ranking such as : $r_{e_1} < \dots < r_{e_n}$ then $\forall i < j, e_i \succ e_j; i, j \in N_+^*$ ⁶ $\implies$ BGS

Each $e_i \in E$ has strict preferences over institutions. $\forall u_h, u_k \in U, u_h \succ_{e_i} u_k$ means that e_i strictly prefers institution u_h over u_k . Each institution $u_k \in U$ has weak preferences over students in E . $\forall e_i, e_j \in E, e_i \succeq_{u_k} e_j$ means that institution u_k weakly prefers student e_i over e_j , allowing for the indifference. $\implies$ BGS. Institutions' preferences are based on students prior university majors. Let denote D the set of available majors; $D = \{d_1, d_2, \dots, d_p\}$. This allows ties between students from the same academic background. However, remark 1 breaks these ties, since there is a strict order in the final ranking. This way, we can express the institutions' strict preference by $e_i \succ_{u_k} e_j$. Since institutions have quotas, the final matching outcome satisfied $|U| \leq |E|$, i.e. $m \leq n$.

Let q_k be the quota of u_k . Since $q_k \geq 1$, the condition $\sum_{k=1}^m q_k \geq |E|$ must be satisfied to assign all students.

In such programs, students are highly qualified professionals and the institutional demand for them is consistently greater than the number of graduates. Moreover, in most cases managers ensure that the number of available positions always exceeds the number of students. $\implies$ BGS. Then, $\sum_{k=1}^m q_k > |E|$.

Unequal-size of sets is not an isolated case to this scenario. In real-world applications like those listed in literature review, equal-sized sets are uncommon. This comes with consequences as Shi et al. (2018b) find that the original stable matching solution is super sensitive for changes in the number of matching members. Since being on the short side of the market confers a large advantage (Ashlagi et al., 2013), students are on the favorite side. This aligns with our model which is student-optimal matching. There is at least one stable matching in which all the members of the smaller set are matched for unequal sizes (McVitie and Wilson, 1970).

To represent the match, we consider a binary variable represented by Biró and McBride (2014) : $x_{ik} \in \{0, 1\}$ for each choice coming from e_i to u_k , as a characteristic function of the matching, where $x_{ik} = 1$ corresponds to the case when e_i is assigned to u_k . Let denote the set of choices by μ . The feasibility of a matching can be ensured with the some constraints : see appendix 6

4.2 Hybrid Matching Algorithm

This student-optimal matching algorithm is based on properties described in the previous section. What distinguishes our algorithm with others is that it includes the remark 1 and institutions' preferences are expressed based on students major instead of e_i directly. A ranking is settled for every e_i within $d \in D$, based on remark 1.

Let define majors of students e_i present in u_k :

$$D_k = \{major(e_i) | (x_{ik} = 1) \in \mu\}$$

The least preferred major for u_k is : $d_l \in \arg \min_{d \in D_k} f_{u_k}(d)$ here $f_{u_k}(d)$ is the preference that u_k assigns to major d .

Among e_i in major d_l , u_k can find the worst final rank:

$$e_j \in \arg \max_{major(e_i)=d_l} r_{e_i} \quad (3)$$

On line 21 of the below pseudocode, we are referring the above e_j

⁶In case of ENA, we assume a strict total order among students because their ranking is based on a large number of assessments, making ties more unlikely.

Algorithm 1: The hybrid algorithm AH-GS

```
1 input :  $n \times m$  Array  $N$  and  $d \times m$  Array  $M$ .  $StuPref$  and  $InstiPref$ 
2 output : a stable matching  $S$ .
3 //Initialization
4 for  $e_i \in E$  do
5    $e_i \leftarrow 0$ 
6 for  $u_k \in U$  do
7    $\text{count}(u_k) \leftarrow 0$ 
8    $\mu \leftarrow \emptyset$ 
9   Start matching by order of students
10 foreach  $i = 1, 2, \dots, n$  do
11   Let  $e_g$  be student to be assign
12    $g \leftarrow i$ 
13   while  $e_g$  not assigned do
14      $u_k \leftarrow \text{next\_pref\_inst}(e_g)$ 
15      $x_{gk} \leftarrow 1$ 
16      $\mu \leftarrow \mu \cup \{x_{gk}\}$ 
17      $\text{count}(u_k) \leftarrow \text{count}(u_k) + 1$ 
18     if  $q_k \geq \text{count}(u_k)$  then
19       Continue
20     else
21       Let  $d_l$  be the least favorite domain of  $u_k$  among other it has
22        $e_j$  is the least pref in  $d_l$  3
23        $x_{jk} \leftarrow 0$ 
24        $\text{count}(u_k) \leftarrow \text{count}(u_k) - 1$ 
25        $\mu \leftarrow \mu / \{x_{jk}\}$ 
26        $g \leftarrow j$ 
27 return  $\mu$ 
```

See the R code [here](#)

4.2.1 Analysis of algorithm 1

Consecutively, each e_i is assigned to his most preferred institution since the algorithm assign even when the quota is reached before removing one student. When a student proposes to a full institution, he is assigned to it and the least ranked in the least preferred major of the institution is removed. The algorithm looks directly for assigning the removed to his next choice. The process continues till when the removed student will be assigned. Then, the algorithm assigns the next student on the ordered list. The process ends when the last student is assigned and the algorithm generates the matching result μ . Majors of students can be replaced by another variable of institution's preferences. In this case, the least preferred student, based on a given preference variable for an institution, will be removed and everything else remains same.

Lemma 1. The output μ of the algorithm 1 is a stable matching with remark 1 constrain

Proof. By contradiction : we assume that there is a blocking pair $(x_{ik} = 1)$ exists in the final matching. This means : $\exists (x_{ik} = 1) \in \mu$ such as

$$\exists e_j \in E \text{ such that } e_j \succ_{u_k} e_i \text{ and} \quad (4)$$

$$\exists u_k \in U \text{ such that } u_h \succ_{e_i} u_k \quad (5)$$

In the algorithm 1, each e_i is assigned first to his most preferred institution; if removed, to the second, etc. The process ends when e_i is assigned to u_k and not removed. This way, e_i cannot be assigned to all $u \in U$ less preferred than his last assignment. This means each $e \in E$ was assigned to all $u \in U$ better than his final assignment and removed. All $u \in U$ which didn't remove e_i and to whom e_i was not once assigned in the process, are less preferred than his current assignment. This contradicts 4 and 5 demonstrating the lemma 1. $\square$

Corollary 4. *The output μ is student-optimal.*

Each $e_i \in E$ who is not assigned to u_k more preferred than his current assignment has been removed from it.

4.2.2 Limitations of Algorithm 1 on incentives

Although incentives exist in the algorithm 1, they are fixed and have limited impact on the outcome.

Lemma 2. *In algorithm 1 the variable X has very limited impact on $f(X, Y)$ compared to Y .*

Proof. Let consider an extreme case where the best student e_1 is the least preferred student of all $u \in U$. In GS algorithm, in such case, a man will be assigned to his last choice woman. At the beginning e_1 is assigned to u_h such as $\forall u_k \in U \setminus \{u_h\}; u_h \succ_{e_1} u_k$. Eventually, when quotas are full, e_1 starts being moved out of his preferred institution in benefit of other students. This will more likely happen since he will always be in the assessed major when it comes to check for the least preferred for an institution. In case of ties in a major, he will remain in the assigned institution. So, e_1 will only stand out either in an institution which is not full or when he shares the same major with at least one student. Moreover in real world applications where grouping like majors and ties do not exist, this algorithm is equivalent to the GS. Even where ties are present, there are uncommon and cannot allow e_1 to plentifully benefit from his first place. $\square$

Lemma 3. *If $|D| = n$, the algorithm 1 is equivalent to the GS; and if $|D| = 1$, then algorithm 1 is equivalent to AH*

Proof. Let reconsider the previous extreme case. If $|D| \rightarrow n$, then there are few ties in major, then $\mu \rightarrow$ GS outcome, then for $x_{1k}, k \rightarrow$ least preferred u by e_1 . If $|D| \rightarrow 1$, then ties are dominant, then $\mu \rightarrow$ AH outcome, then for $x_{1k}, k \rightarrow$ most preferred u by e_1 . $\square$

Corollary 5. *There is no policy variable in algorithm 1*

The order ranking and preferences of both sides are known. The planner does not have any strategic role, since there is no way to optimize inputs X and Y . Sure, he can strategically work on grouping of majors to influence the algorithm but this is limited as it should be fixed in many cases.

4.3 Hybrid Matching Algorithm with policy variable

Limitations of the previous algorithm make relevant the definition of a control variable which allows the planner to determine the contribution of each .

4.3.1 The policy control variable

Let v be the policy variable representing the trade-off between X and Y . From previous developments, we know that v has positive relationship with one variable and negative with another. To ensure that the CES function is homothetic (i.e., has constant returns to scale or simplifies share allocation) and homogeneous, we assume $\alpha + \beta = 1$. Then, we define:

$$f(X, Y) = (\alpha X^\rho + (1 - \alpha)Y^\rho)^{\frac{1}{\rho}} \quad \text{with } \rho \in (0, 1), \text{ and } \beta = 1 - \alpha.$$

Lemma 4. *In the function $f(X, Y)$, α serves as a proxy for the policy variable v , capturing the flexibility between incentives and preference matching, such that $\alpha = \phi(v)$.*

Proof. The lemma 3 tells us that the number of students major is the perfect proxy for the policy variable since it allows to move flexibly from AH to GS. However, that variable is exogenous. We retain only two extremes in lemma 3 : $[1, n]$ as characteristics of v so that $v \in [1, n]$. We know that $\alpha \in [0, 1]$. We define a function $\alpha(v)$ such that:

$$\lim_{v \rightarrow 1} \alpha(v) = 0 \quad \text{and} \quad \lim_{v \rightarrow n} \alpha(v) = 1.$$

A natural and unique linear transformation satisfying these conditions is:

$$\alpha = \frac{v - 1}{n - 1} \quad (\text{Logically the number of students should be higher than 1})$$

The policy variable v affects the parameter α through a differentiable and strictly increasing function, i.e., $\alpha = \phi(v)$ with $\phi'(v) > 0$. Let define the contributions inside the CES aggregator as: $C_X = \frac{v-1}{n-1}X^\rho$, $C_Y = \frac{n-v}{n-1}Y^\rho$, so that: $f(X, Y; \alpha) = (C_X + C_Y)^{1/\rho}$.

First, let's check the relationship between the policy v and variables X and Y . We aim to prove that: if v increases, the contribution C_X increases and C_Y decreases. If v decreases, the contribution C_X decreases and C_Y increases. The two contributions should move in opposite directions in reaction to changes in the policy variable v .

$$\frac{dC_X}{dv} = \frac{d(\alpha X^\rho)}{dv} = X^\rho \cdot \frac{d\alpha}{dv} = \frac{1}{n-1}X^\rho > 0,$$

since $X^\rho > 0$ and $\frac{1}{n-1} > 0$. Therefore, C_X increases with v .

$$\text{Similarly, } \frac{dC_Y}{dv} = \frac{d((\frac{n-v}{n-1})Y^\rho)}{dv} = -\frac{1}{n-1}Y^\rho < 0,$$

since $Y^\rho > 0$ and $\frac{1}{n-1} > 0$. Therefore, C_Y decreases with v . This proves that as the policy variable v increases, the contribution of one variable (X) increases while the contribution of the other (Y) decreases. Conversely, when v decreases, the opposite occurs. Thus, the policy v has effects of opposite sign on the contributions of X and Y in the CES function. While α is treated as a policy variable under the planner's control, the inputs X and Y are considered exogenous variables representing candidate performance and institutional quality, respectively. Their values are not fixed but result from underlying processes (such as assignment quality) that lie outside the planner's immediate decision in the CES framework.

Corollary 6. *The policy is more effective in presence of few students.*

$\lim_{n \rightarrow 1} \frac{dC_X}{dv} = +\infty$ (that's the highest contribution) and $\lim_{n \rightarrow 1} \frac{dC_Y}{dv} = -\infty$ (Then, the minimum contribution which is 0). This means that near $n = 1$, a little change in the policy variable yields in extreme marginal outcomes). Similarly, When $n \rightarrow \infty$, all margins contributions are reduced.

Second, the property of convexity stands.

$\alpha(v) = \frac{v-1}{n-1} \in [0, 1]$ is a strictly increasing function mapping $v \in [1, n]$ to $\alpha \in [0, 1]$. We evaluate the function at the boundaries:

- At $v = 1$, we have $\alpha = 0$, hence: $f(1) = (Y^\rho)^{1/\rho} = Y$.
- At $v = n$, we have $\alpha = 1$, hence: $f(n) = (X^\rho)^{1/\rho} = X$.

Now consider an interior point $v^* \in (1, n)$ such that $\alpha = \frac{v^*-1}{n-1} \in (0, 1)$. Then: $f(v^*) = (\alpha X^\rho + (1-\alpha)Y^\rho)^{1/\rho}$.

Since $\rho \in (0, 1)$, the CES function is strictly concave. By convexity's property, we have: $f(v^*) > \max\{f(1), f(n)\}$. Then, the maximum of $f(v)$ is not achieved near the boundaries $v = 1$ or $v = n$, but at some interior point $v^* \in (1, n)$. Since the indifference curve is convex with diminishing marginal rate of substitution, the endpoints of the policy variable v do not yield a maximum outcome for the function. However, in our model, the uncontrolled part is stochastic and not necessarily zero, making higher outcomes near the boundaries possible, but less probable. $\square$

4.3.2 Definition

From Corollary 6, we know that v depends on n which is exogenous. To represent v in algorithm, let's create equal classes in the final ordered ranking. We consider a set of Classes $C = \{c_1, c_2, \dots, c_l\}$ with $|C|$ the cardinality of C representing the number of classes. Let recall that $n \in \mathbb{N}_+^*$ is the number of students. Then $\frac{n}{|C|}$ is the number of elements of each $c \in C$. Since there is an order between sets and elements of each set; the first class is made of $\frac{n}{|C|}$ first e_i on the final ranking. The following sets have the same logic; each class contains $\frac{n}{|C|}$ ordered elements starting from the best ranked who is not in the previous class. So we can write the rank of classes such as such as : $r_{c_1} < r_{c_2} < \dots < r_{c_l}$ then $\forall s < t, c_s \succ c_t; s, t \in [1, l]$. There is no intersection between two sets.

Remark 2. if $c_s \succ c_t$; each $e_i \in c_s$ has priority on each $e_j \in c_t$.

This is the core property of the current development. Regardless of the preferences of both sides, each student has priority over all other in classes less ranked than his current.

Lemma 5. In the algorithm, the class' size $\frac{n}{|C|}$ is a relevant proxy of the policy variable v . Since n is constant, the number of classes $|C|$ is the variable.

Proof. We know that $v \in [1, n]$. Let's consider endpoints : if $|C|$ decreases and tend to the boundary 1, then $\frac{n}{|C|} \rightarrow n$. The class' size tend to the maximum value. So, classes (subsequently the remark 2) tend to disappear and the scenario tend to the GS model which maximizes the Y disfavoring X . If the number of classes $|C|$ increases and tend to the extreme n , then $\frac{n}{|C|} \rightarrow 1$. The class' size tend to the minimum value. So, each e_i tend to have his own class. In this case the remark 2 will allow each e_i to have priority over all other less ranked than him leading to AH model. The model maximizes X disfavoring Y . This confirms the opposite relationship of v on X and Y proved in lemma 5. Then $\frac{n}{|C|} \equiv v$. v is the size of each class; $C = \{c_1, c_2, \dots, c_{\frac{n}{v}}\}$. The planner can lead the policy by defining the size of sets. $\forall c_s \succ c_t$, each $e_i \in c_s$ has more options than each $e_i \in c_t$. $\square$

Theorem 3. AH-GS model with v consists of previous AH-GS model in each class $c \in C$.

Proof. Let consider $|C| = 3$, then $C = \{c_1, c_2, c_3\}$. Thanks to the remark 2, each $e_i \in c_2$ has priority on each $e_i \in c_3$; i.e. $e_i \in c_2$ cannot lose his preferred assignment in favor of $e_i \in c_3$ regardless of both sides preferences. For the assignment, each $e_i \in c_2$ is competing with his peers and with $e_i \in c_1$. However, since $c_1 \succ c_2$ each $e_i \in c_1$ will be prioritized by remark 2 over each $e_i \in c_2$. Then, the matching process will happen within the class $e_i \in c_2$. Each class excludes the worst ranked than him and he is excluded by the best ranked than him; and remains alone for the matching process. $\square$

The algorithm AH-GS with v will consist of adding new features including the v policy which appears in the existing loop and the new one.

See the algorithm 2 on next page

4.3.3 Analysis of algorithm 2

The logic of the algorithm 1 stands in addition to some key points. First, the algorithm selects the first class c_1 . Each student in c_1 is assigned to his most preferred institution. When a student proposes to a full institution, he is assigned to it. Then, the removing process is applied only within the first class. The algorithm looks directly for assigning the removed to his next choice. The process continues till when the removed student is assigned. Afterward, the algorithm assigns the next student on the ordered list within c_1 . The same process continues in all classes. The key point relies on the matching in the same class. When the last student is assigned, the algorithm generates the matching result μ . This algorithm can be adapted to a scenario without majors and ties by removing the "least preferred" e_j logic.

Theorem 4. The outcome X and Y in AH-GS model tend respectively to the maximum X in AH and maximum Y in GS. Thanks to theorem 2, the propriety (6) is verified for the AH-GS model.

Proof. AH-GS model with v policy is discriminatory, penalizing deeply least ranked students against their preferences. The more a class tend to the first place, the less Y is maximized within it. In c_1 there is v students for all available places in institutions. In addition to the ranking advantage (which is limited within a class, see lemma 2), each $e_i \in c_1$ has a compelling advantage to choose any institution without being threatened by any student of other classes. The equilibrium is now sustained by those ranked lowest, who bear a heavy cost. When the last class is proceeded, few places remain and choices are more oriented by the relevance of the match major-training. Then, it is definitely preferable for each student to be on top. Thus, the incentive is furnished and the variable X tend to its maximum. The variable Y is mostly sustained by least ranked students and it still influences matching in top classes though with less impact. $\square$

Algorithm 2: AH-GS Algorithm with v as the policy variable

```
1 input :  $n \times m$  Array  $N$  and  $d \times m$  Array  $M.StuPref$  and  $InstiPref, v.ClassSize$ 
2 output : a stable matching  $S$ .
3 //Initialization
4 for  $e \in E$  do
5    $e_i \leftarrow 0$ 
6 for  $c \in C$  do
7    $c_s \leftarrow 0$ 
8 for  $u \in U$  do
9    $count(u_k) \leftarrow 0$ 
10   $\mu \leftarrow \emptyset$ 
11  Start matching by order of students
12 foreach  $s = 1, 2, \dots, \frac{n}{v}$  do
13   foreach  $i = 1, 2, \dots, v$  do
14     Let be  $e_g$  be student to be assign
15      $g \leftarrow i$ 
16     while  $e_g$  not assigned do
17        $u_k \leftarrow next\_pref\_inst(e_g)$ 
18        $x_{gk} \leftarrow 1$ 
19        $\mu \leftarrow \mu \cup \{x_{gk}\}$ 
20        $count(u_k) \leftarrow count(u_k) + 1$ 
21       if  $q_k \geq count(u_k)$  then
22         Continue
23       else
24         if  $\nexists e_j \in (C_s \cap U_k)$  then
25            $x_{gk} \leftarrow 0$ 
26            $count(u_k) \leftarrow count(u_k) - 1$ 
27            $\mu \leftarrow \mu / \{x_{gk}\}$ 
28         else
29            $\forall e_j \in (C_s \cap U_k)$ :
30             Let  $d_l$  be the least favorite major of  $u_k$ 
31              $e_j$  is the least pref in  $d_l$  3
32              $x_{jk} \leftarrow 0$ 
33              $count(u_k) \leftarrow count(u_k) - 1$ 
34              $\mu \leftarrow \mu / \{x_{jk}\}$ 
35              $g \leftarrow j$ 
36   end
37 return  $\mu$ 
```

5 Applications

Previous sections were developed along ENA assignment application which is a type of college assignment scenario. This application type is currently over-leading as employers are considering important for their employees to undergo mid-term or long term training after the recruitment before assigning them to different departments. Considering the AH-GS model, employers are no longer limited to AH, GS or to improvise a composite and complex system missing a flexible middle-ground solution. Applications of AH-GS model in domains with same characteristics with ENA are countless : in public sector (INSP-France, ENA-DR Congo, Foreign Service Institute and Federal Bureau of Investigation-USA, National Administrative training-India, National Public Service-Japan, etc.); Private Sector (McKinsey & Company, Boston Consulting Group, Bain, Goldman Sachs, JPMorgan Chase, Google, Microsoft, Amazon, Airlines companies, etc.), Healthcare system (Medical Residence in the USA, the UK, Canada, etc.), International Organization (World Health Organisation, United Nation Fellowship Programs, etc.). Re-

ardless of their type, many organizations have adopted that transitional training particularly for their cutting-edge departments. Since that type of application has been developed throughout the paper, we consider only the military assignment scenario before presenting applications in other domains.

5.1 Army assignment

In military context, particularly in specialized units, a ranking system is used to assign personnel. One relevant example is the U.S. Army Officer candidate School (OCS) which is in charge of training, educating and commissioning officers. The recruitment policy to the OCS (army, 2025) and the assignment policy in infantry (army, 2021) ensure that the process is clear and transparent to everyone. Despite the complexity of the army’s talent alignment process, its results closely resemble those of the Gale–Shapley optimal matching, suggesting that an AH–GS perspective can provide valuable insights. The same applies to the Marine Corps’ assignment system (army, 2013) at the Basic School, which, as currently designed, does not maximize the previously defined bivariate function, since it considers only a single dimension of the matching process.

Some assignments systems consider many factors, which can be assimilated to variables X and Y , to assign candidates. This is the case for the Singapore Armed Forces Training Institute. Official informations⁷ list variables included in the assignment process : military performance, commander’s recommendation, organizational requirements, education qualifications and medical fitness. The two first elements should be assimilated to variable X and three others to variable Y . From there, the framework developed can be applied to allow policy flexibility between variables X and Y to optimize the outcome.

5.2 Inclusive development strategies

This application is developed within a realistic development framework that emphasizes practical relevance and empirical consistency. It responds to longstanding concerns in the literature about the gap between economic theory and real-world applicability. As highlighted by Romer (2016), much of modern economics has been described as “post-real,” while several scholars have drawn attention to the limitations of overly abstract or idealized models (Coase, 1993; Kirman, 2010). Others have stressed the need for approaches that better inform policy decisions (Keen, 2011; Lawson, 1997) and incorporate key features of real economies (Stiglitz, 2018). In this spirit, and acknowledging such critiques (Von Hayek, 1989), the following section presents a straightforward and policy-oriented application designed to illustrate the practical dimension of the proposed framework.

In the Solow (1956) growth model, factors of production are assumed to be flexible and freely substitutable. In practice, however, this assumption is often violated due to institutional and regulatory constraints. Empirical evidence highlights several imperfections, such as onerous regulations and labor laws (McKenzie, 2017), short-term rigidities in the labor market (Vooren et al., 2019), and quasi-fixity of labor (Oi, 1983).

When a central government (CG) seeks to maximize national output through policies affecting capital and labor, it may face such frictions, particularly in multi-level governance systems. For instance, in federal structures (e.g., Germany, Canada, Belgium, India, and the United States), the CG typically manages capital-related policies (national investment, macroeconomic policy, infrastructure), whereas subnational governments hold authority over labor regulations.

This institutional separation can create coordination inefficiencies—especially when the most effective labor policies fall outside the CG’s direct control. In Canada, for example, the Infrastructure Bank oversees national investment programs (Investment policy, 2024), and the Department of Finance manages fiscal and macroeconomic policy (fiscal and tax policy, 2022), while provinces independently define employment standards such as minimum wage, statutory holidays, and termination conditions. A similar situation is observed in India where different layers of governance are Union government, State governments and local governments enhancing the possibilities of this constrain.

The modeling of these scenarios can be applied in one specific sector. In the DRC, the Constitution (2006) which is the highest law, clearly assigns exclusive responsibilities to the central government (Article 202) and to the provinces (Article 204). For example, to maximize the national education outcome in the DRC, the planner has to work on the development of university level and High school level. The

⁷<https://www.cmpb.gov.sg/life-in-ns/saf/after-basic-training>

below function can represent that scenario :

$$\max_{U,E} Edu(U, E) = (\alpha U^\rho + \beta E^\rho)^{\frac{1}{\rho}}$$

Where : U is the number of universities built or funded by the central government (Art. 202.22); E is the number of primary and secondary schools built by provinces (Art. 204.13); α, β represent weights or productivity coefficients, ρ is the elasticity parameter. The appendix 6 presents the construction of such model in the fields of public health, agriculture, epidemic control, and infrastructure development, each based on the respective DRC constitutional mandates. These situations have prompted economists to design coordination policies aimed at maximizing such joint outcomes. In practice, however, each level of government often pursues its own political or electoral priorities, thereby weakening the effectiveness of coordination mechanisms and reducing the overall efficiency of public policy.

Let's present some inclusive development strategies to introduce how the AH-GS model can address this constrain. The theory of balanced growth states that a country should invest in different sectors to promote a dynamic diversification of the economy, thus ensure a long term stability (Bass, 2009). While interacting with international organizations like the international monetary fund (IMF), developing countries are encouraged to promote balanced growth (IMF, 2025; Ahmad et al., 2003; UNDP, 2024). For the IMF (2024) the unbalanced growth explains why the Chinese economy has started slowing. The TPA (see illustration in appendix 6) presented in literature, aligns with the IMF proposal to adopt structural reforms and rebalancing China's growth towards a more balanced, inclusive, and green growth. In the same context the development theory of big push wants countries to invest simultaneously in many sectors of the economy to lead quickly the development (Murphy et al., 1989). Sectors to be financed need to exhibit increasing returns in order to trigger development dynamics (Kattel et al., 2009).

To achieve balanced long-run growth, economic development must be inclusive, addressing the needs of poorer regions and marginalized populations. When the central government (CG) controls only capital-related policies, it should therefore prioritize investments in the least-developed sectors and subnational regions seeking a match between regional needs and available projects. This logic is comparable to China's TPA program, where the assignment process aims to optimally match regional needs with public investment projects. In such a setting, the labor contribution to national output can be considered stochastic, as subnational governments may follow heterogeneous priorities that are not perfectly aligned with national objectives. This is precisely where the AH-GS model can intervene. Beyond the capital variable X , the CG can influence the effective labor input Y indirectly by designing incentive-compatible assignment mechanisms linking project allocation to measurable labor indicators that reflect inclusive growth targets. Through the AH-GS framework, the CG maintains control over capital allocation (X) while guiding the effective deployment of labor efforts (Y) via incentive mechanisms defined by a policy variable v . Under such coordination, the resulting aggregate outcome function analogous to $f(X, Y; v)$ is expected to yield higher national performance compared to uncoordinated or purely performance-based systems, as demonstrated throughout this paper. The same logic can be applied to sectoral models, such as those developed for the DRC's education, health, and infrastructure sectors (see Appendix 6).

5.3 Public health policy

This section is not a straight application since this may require deep studies involving algorithmists, gametheorists and health policy planners to explore applications of both incentive and matching in health system. Rather, we present how health system is a good ground for AH-GS model.

Incentives have important place in health care policies to influence behavior, improve efficiency, and achieve better health outcomes. Different studies (Michaud et al., 2022; OECD, 2023) have proved how incentives are effective for health policies. Though the financial motivation is the most frequent, the WHO (2000) states that both monetary and non-monetary incentives have to be combined for higher outcome. However when it comes to list non-monetary incentives, the WHO does not include a reward in a given matching process. Whereas, the health system is known as the one with a lot applications in matching process and contest. For example, A. Roth received the Nobel prize for his work on market design for kidney donors involving a matching process which has saved a lot of lives. Moreover, current monetary incentives are based on pay-for-Performance system in which beneficiaries are compared with one another based on a set of performance indicators and those who achieve a high level of performance are rewarded financially (Abduljawad and Al-Assaf, 2011). Thereby, the two variables X and Y of the AH-GS model are present in health policy. Additionally, it is very interesting to realize how the idea of uncontrolled variable developed in the previous section might be more relevant in health policy.

For example the function of health outcome (eg. life expectancy, etc.) $H = (P, G)$, with P : public health spending; G : genetic predisposition of population (not controlled). The government can invest in hospitals or prevention but can't change people's genes. However the government can invest to leverage P to reduce the harm caused by G .

5.4 Maximization of Sport Team Revenue

Let us consider a sport team whose owner seeks to maximize total revenue from two primary sources : performance-based rewards from victories in competitions and future income from selling players whose market value increases after strong performances. In this setting, both proven veterans and promising young talents are available, and all participate in a short training camp before the season begins. The number of competition spots is limited, requiring an internal selection process.

If the team selects players solely on the basis of proven past records, prioritizing those with extensive prior achievements, the process reflects the logic of the GS model : each competition spot is matched with the most experienced available players, and players will be selected regardless of training performance. While this approach reduces uncertainty by relying on established track records, it may restrict opportunities for unproven but potentially high-performing players. Moreover, since the selections are known in advance, internal competition during training is reduced, which can limit performance improvements across the squad. If, instead, the team selects only the best performers during the training camp, without considering prior records, the process corresponds to the AH model. Here, past achievements are disregarded, and selection depends exclusively on short-term performance indicators. This method can identify new talent and stimulate competition during training, raising the performance level of all players. However, it carries the risk that exceptional training performance may not translate into sustained results in competitive matches.

The hybrid AH-GS model integrates these two approaches by considering both training performance and prior records in the selection process. In practice, this means that experienced players with consistently high performance are favored, while emerging talents demonstrating exceptional ability during training are also given opportunities. This balanced method not only increases revenue through the discovery and development of new players but also sustains a high level of internal competition, which improves the overall performance of the team. Thereby, both chances of earning trophies and selling expensive players are improved. By applying the hybrid model, team owners and managers are provided with a structured framework that reduces reliance on subjective criteria, addresses the limitations inherent in using only AH or GS, and aligns selection decisions with revenue-maximization objectives. In this configuration, AH-GS retains the main advantages of both methods (the reliability of GS and the competitive stimulus of AH) while avoiding their principal drawbacks (the reduced competition of GS and the performance uncertainty of AH).

6 Discussion and conclusion

Some decision-makers had already intuitively realized the need to combine the AH method with the GS algorithm. Some adopted mixed approaches, such as allowing the top n candidates to choose their placements freely (as in AH), while assigning the remaining candidates using the GS algorithm to optimize skill matching. However, such methods introduced significant subjectivity and injustice, as the process abruptly ends participation for those not among the first n , leading to inequitable outcomes.

Although many studies explore incentives and extensions of the GS algorithm, the interaction between AH and GS models remains relatively underexplored. It is true that some large organizations employ hybrid assignment algorithms that consider multiple factors. Yet, upon closer inspection, some of these approaches refer to a single variable or produce results equivalent to the men-optimal GS solution. Others lack a clear rational framework, leaving room for substantial subjectivity.

In this paper, we have developed theoretical models, constructed algorithms, and presented applications that provide empirical evidence. The objective is not to suggest the replacement of others models, but rather to offer decision-makers an additional policy option; one that is clear, concise, and flexible. After evaluation, a decision-maker may determine that the hybrid function is not well suited to their specific context and may instead return to a pure AH or GS model. What matters is that the choice is made from a structured and informed perspective.

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Appendix A : The China Targeted Poverty Alleviation

Y. Yang and Liu (2021) have represented the core matching system of TPA. First, deep studies are conducted in every region to determine root causes of poverty. Afterward, tailored projects are assigned to each region to boost inclusive growth.

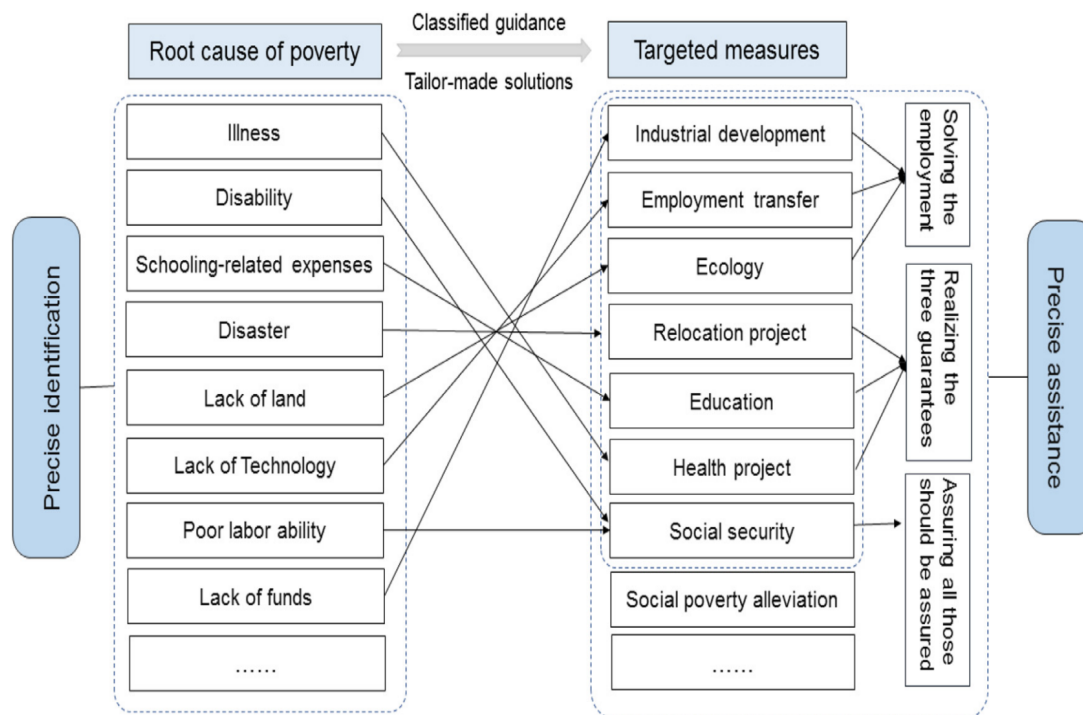


Figure 1: Match between root cause of poverty and targeted measures

Appendix B : Conditions for a match

To verify the feasibility conditions of our model, we consider a binary variable represented by [Biró and McBride \(2014\)](#). $x_{ik} \in \{0, 1\}$ for each choice coming from e_i to u_k , as a characteristic function of the matching, where $x_{ik} = 1$ corresponds to the case when e_i is assigned to u_k . Let denote the set of choices by μ . The feasibility of a matching can be ensured with the following two sets of constraints:

$$\sum_{k:(e_i, u_k) \in \mu} x_{ik} \leq 1 \text{ for each } e_i \in E \quad (6)$$

$$\sum_{i:(e_i, u_k) \in \mu} x_{ik} \leq q_k \text{ for each } u_k \in U \quad (7)$$

Here, (6) implies that no student can be assigned to more than one institution, and (7) implies that the upper quotas of the institutions are respected.

Appendix C : strict quasi-concavity

From the CES function : $f(X, Y) = (aX^p + bY^p)^{1/p}$ where $a > 0$, $b > 0$, and $0 < p < 1$.

Case 1: Only one variable is used

Let $y = 0$. Then the function reduces to: $f(X, 0) = (aX^p)^{1/p} = a^{1/p} X$

Case 2: Both variables are used

Assume $X > 0$ and $Y > 0$. Then: $f(X, Y) = (aX^p + bY^p)^{1/p}$. Factor out aX^p :

$$f(X, Y) = \left(aX^p \left(1 + \frac{b}{a} \left(\frac{Y}{X} \right)^p \right) \right)^{1/p} = a^{1/p} X \left(1 + \frac{b}{a} \left(\frac{Y}{X} \right)^p \right)^{1/p}$$

$$f(X, Y) = f(X, 0) \cdot \left(1 + \frac{b}{a} \left(\frac{Y}{X} \right)^p \right)^{1/p}$$

Since $\left(\frac{Y}{X} \right)^p > 0$ and $\frac{b}{a} > 0$, it follows that: $\left(1 + \frac{b}{a} \left(\frac{Y}{X} \right)^p \right)^{1/p} > 1$. Thus, $f(X, Y) > f(X, 0)$

Therefore, for $0 < p < 1$, the CES function satisfies:

$f(X, Y) > \max \{f(X, 0), f(0, Y)\}$ for all $X > 0, Y > 0$.

Appendix D : Modeling joint competencies of central and provincial Governments based on DRC highest law : examples

1) Public health

Objective: Minimize avoidable mortality through central hospital investment and provincial health-care services.

$$\min_{H, P} Mortality(H, P) = \frac{\gamma}{H^\theta \cdot P^{1-\theta}}$$

Where: H is the national planning, hospitals and medical infrastructure (Art. 202.36.h); P is the provincial healthcare services (Art. 204.18); γ is the mortality scaling constant; θ is the elasticity parameter.

2) Agricultural development

Objective: Maximize agricultural production through national programs and provincial implementation.

$$\max_{A_n, A_p} Agr(A_n, A_p) = \delta_1 \cdot A_n + \delta_2 \cdot A_p - C(A_n, A_p)$$

Where: A_n represent agricultural programs at the national level (Art. 202.25); A_p represent execution of agricultural programs at the provincial level (Art. 204.20); C is Joint cost function; δ_1, δ_2 : are productivity parameters

3) **Epidemic control**

Objective: Minimize epidemic spread through joint medical infrastructure and vaccination efforts.

$$\min_{R,V} Infection(R,V) = \frac{\lambda}{R \cdot V}$$

Where: R represent the national laboratory systems and technical coordination (Art. 202.36.h); V is the variable for provincial vaccination and epidemic control campaigns (Art. 204.18, 204.22) and λ is the constant of proportionality

4) **Infrastructure development**

Objective: Maximize national logistics and connectivity through central and provincial infrastructure investments.

$$\max_{T_n, T_p} Infra(T_n, T_p) = \eta \cdot \log(T_n + T_p)$$

Where: T_n is investment in national transport infrastructure (Art. 202.34); T_p represent provincial roads and local public works (Art. 204.11, 204.24); η is the return on synergy from joint infrastructure.